Move two S3 buckets from Sydney to Oregon

Approach

I use **AWS DataSync (Basic mode)** to copy each Sydney bucket to a new Oregon bucket. The data moves S3 to S3 over the AWS backbone, with no agent and no servers to manage. DataSync handles retries, checksum verification and per-object reporting, so I don't have to build those myself.

Estimated one-time cost: about $62. $49 of that is inter-region transfer, which every AWS-native method pays. The cheapest option, a plain AWS S3 sync, would cost about $57.

Why DataSync and not the cheapest option

I read "as cheaply as possible" to include engineering time, not only the AWS bill. DataSync costs about $5 more than a manual sync. For that, it gives me built-in integrity checks, automatic retries and an auditable task report. Writing and running my own verification would cost more than $5 of engineer time. If the brief strictly means the lowest AWS bill, aws s3 sync is the answer.

I chose **Basic mode** over Enhanced mode. 500,000 objects is well within Basic mode's limits, and Enhanced mode adds $0.0025/GB plus $0.55 per task execution for speed I don't need.

Assumptions

- 500 GB total, about 500,000 objects, all in S3 Standard, encrypted with SSE-S3.
- Same AWS account. Only current object versions need to move, and versioning is off.
- Writes can be frozen briefly for a final incremental run.
- New bucket names (names are global, so the Sydney names can't be reused until those buckets are deleted).
- On-Demand list prices in USD, checked October 2026.

Steps

1. Create the two destination buckets in us-west-2 with Block Public Access on and default encryption.
2. In the DataSync console in **US West (Oregon)**, create one task per bucket pair:
3. Source location: the Sydney bucket. Destination location: the Oregon bucket, storage class Standard.
    - Task mode: **Basic**. Transfer mode: **only data that has changed**. Verification: **only verify data transferred**.
    - Keep deleted files, preserve object tags, task report level **successes and errors**.
    - Let DataSync create the IAM role for bucket access.
4. Run both tasks. Freeze writes, run each task once more to pick up changes, then verify.

5. Cut the application over, keep Sydney for 7 days, then delete it.

Cost breakdown

Verified in the AWS Pricing Calculator

I entered the same inputs in the AWS Pricing Calculator: shared estimate.

Calculator line	Region	Inputs	Monthly cost shown
AWS DataSync	US West (Oregon)	Basic mode, 500 GB	$6.25
AWS Data Transfer	Asia Pacific (Sydney)	Outbound to other regions, 500 GB	$49.00
Amazon S3	US West (Oregon)	500 GB Standard, 503,000 PUT/LIST, 1,000,000 GET	$14.41
Amazon S3	Asia Pacific (Sydney)	117 GB Standard, 2,000 PUT/LIST, 1,000,000 GET	$3.38
Total			**$73.04**

The calculator treats every line as monthly, so its total includes one month of Oregon storage. One-time migration cost = $73.04 − $11.50 = **$61.54**. Its "12 months" figure assumes the migration repeats every month and doesn't apply here.

Alternatives considered

Option	Cost vs DataSync	Verdict
DataSync Basic (chosen)		Managed, verified, reported.
aws s3 sync	About $5 less	Cheapest, but I'd have to build and run verification myself.
DataSync Enhanced	About $2 more	Faster, built for tens of millions of objects. Not needed at 500,000.

Verification

I use three free layers. Each one catches something the others can miss.

1. DataSync verification and task report (content, per object). With "only verify data transferred", DataSync checksums every object during the copy and checks it again at the destination. Any mismatch fails the task. The task report lists every object with its transfer and verification status, so I confirm zero failures and zero skipped objects.

2. Count and size check right after the copy (completeness). DataSync only grades what it attempted. If a filter, permission or wrong prefix made it skip objects, it can still report success. So I compare all four buckets directly:

bash
```
aws s3 ls s3://BUCKET --recursive --summarize | tail -2
```

Total objects and total size must match exactly between each source and destination pair.

3. CloudWatch storage metrics (independent confirmation). S3 publishes NumberOfObjects and BucketSizeBytes for every bucket once a day at no cost. During the 7-day soak, I compare each source and destination pair in the CloudWatch console (namespace AWS/S3):

Metric	Dimensions	Expected
NumberOfObjects	StorageType = AllStorageTypes	Equal for each pair
BucketSizeBytes	StorageType = StandardStorage	Equal for each pair

These metrics lag by about a day, and a new bucket can take a day or two to show its first data point, so I use them to confirm, never as the go/no-go check.

Sources (checked October 24, 2026)

- S3 pricing: https://aws.amazon.com/s3/pricing/